

Element D

D1: Process for Generating & Comparing Possible Solution

Morphological Chart: The Morphological Chart described below was made with the goal to gather our design ideas that we believe could potentially meet the design requirements laid out above. The goal through this chart was to be able to use this organized chart to piece design ideas so that we can satisfy as many design requirements as possible.

Functions/ Concepts Importance Rating (1=Most Important)	Gravity Slope Design	Concept 2	Siphon Design	Brita Design
Produce sufficient water flow (1)	2in Diameter Circle cut into tank. Water flows by force of gravity.	Use of an electric water pump.	Siphon system	Traditional 2-way or 3-way valve to allow or not allow water flow
Portability. (2)	Handlebars	Stand with Wheels.	Carried by the bottom of sink (No add-on)	Shoulder Strap (To be carried on a shoulder)
Student Functionality (3)	Interchangeable sink handles	Instruction label as to which handle does which	Labeling of Each Piece with the name (as well as possible numbering)	List of instructions and motion/pictures on how to operate the different parts of the sink
Standard Sink appearance (4)	Metal water basin + Traditional Faucet handles (L shape) J shaped traditional faucet	3D Printed sink bowl + button to start water flow+3D printed faucet	Plastic Container with Metal water basin	A visible sensor that detects hand movement, to signal the water to begin flowing
Safety(5)	Rounded off edges	Plastic casing around sink (During Storage	Warning Sign of potential hazards	Child latches, to prevent children from taking parts apart.
Cleanliness(6)	Use of materials that are easy to be cleaned with external products (acrylics, and	Inclusion of Soap container	Inclusion of Wet wipes (to wipe down sink)	A mini date-sheet, to remind the teachers of the last time cleaned, and approximately the

	wood, etc)			next time the sink needs to be cleaned
Storage(7)	Top that covers the sink bowl and faucets as well as the water tank.	Ability to Disconnect water tank, sink bowl, and waste tank to individually store	Hinges to disconnect pieces, and fold pieces.	No modification to enhance storage ability
Customizability(8)	Interchangeable sink handles	Interchangeable mirror	Magnetic or velcro capabilities	Ability to adjust flow rate

Scoring Rubric for Design Requirements

Design Objectives	Excellent (4)	Good (3)	Satisfactory (2)	Unacceptable (1)
Produce sufficient water flow (1)	>1 Gallons of Water per Minute	.5 gallons of water per minute	.25 gallons per minute	<.125 gallons per minute
Portability. (2)	<=50lbs	50-55lbs	56-60 lbs	>60lbs
Student Functionality (3)	100% of students are able to manipulate the sink	99%-80% of students are able to manipulate the sink	79%-60% of students are able to manipulate the sink	59%-0% of students are able to manipulate the sink
Standard Sink appearance (4)	Rating of 10	Rating of 8-9	Rating of 5-7	Rating of 1-4
Safety(5)	0 edges or ridges identified	1-2 edges or ridges identified	3-4 edges or ridges identified	5 or more edges or ridges identified
Cleanliness(6)	No visible smears or liquids that were left on sink after a week of use	1-3 smears or liquids that were left on sink after a week of use	4-5 smears or liquids that were left on sink after a week of use	5 or more smears or liquids that were left on sink after a week of use

Storage(7)	70 ft cubed or less	85 ft cubed	100 ft cubed	115 ft. cubed or more
Customizability(8)	3 or more parts of the sink can be modified for student/teacher use	2 parts of the sink can be modified for the student/teacher use	1 part of the sink can be modified for the student/teacher use	No parts of the sink can be modified for the student/teacher use

D2: Design Solution Choice

The following Decision Matrix is developed based on design requirements, as well as tangible mathematical score ranges shown above in the Design Objectives chart. These scores are all based on the perceived capability of these sink designs through the conversations of our group as well as the conversations between our supervisors, including Drexel Professor's . These prototypes were not all built and scored through experimentation due to project costs and constraints. Further explanations for each final score are listed below.

				Sink Decision Matrix			
				rated from 1-4			
	6	Brita Design		Siphon Design		Gravity Slope Design	
Criteria	Weight	Rating	Values	Rating	Values	Rating	Values
Produce sufficient water flow	10%	3	0.30	4	0.40	4	0.40
Portability.	20%	3	0.60	3	0.60	1	0.20
Student Functionality	20%	4	0.80	1	0.20	4	0.80
Standard Sink appearance	15%	3	0.45	2	0.30	3	0.45
Safety	10%	3	0.30	3	0.30	2	0.20
Cleanliness	10%	2	0.20	4	0.40	2	0.20
Storage	5%	4	0.20	3	0.15	1	0.05
Customizability	10%	2	0.20t	2	0.20	2	0.20
Total:	100%		3.05		2.55		2.50

Design Reflections

Brita Design

When looking above, it was clear that the Brita Design was the highest scoring. The explanation for each score for each criteria is below

- **Produce sufficient water flow (3)**- Due to the large volume of water each tank can hold ,(2.5) gallons of water, we are confident that the flow rate will not be an issue for the students and teachers. Also, due to the minimal amount of piping from tank to faucet, there's minimal concern that the flow of gravity will not sufficiently allow for the water to flow 4 gallons of water a minute if needed.
- **Portability(3)**- Due to the hollowness of the container, most of the weight should be allocated to the water tanks and sink bowl. For that reason based on materials researched, the sink could be built under the weight of 50lbs.
- **Student Functionality(4)**- While the final rating is unknown, we believe due to the purchase of a sink bowl, as well as 3D printed faucet handles, will allow for all students to understand, or at the very least be able to manually adjust the faucet handle with the help of the teacher.
- **Standard Sink Appearance(3)**- Again, while the final rating is unknown since this prototype was yet to be built during the phase of the decision matrix. However, due to the purchase of a traditional sink bowl, we believe this allows for this prototype to appear as a traditional sink that students may be able to
- **Safety- (3)** It's likely that the sink container won't have any sharp edges that will be present once the product is sent to the stakeholders for daily use.
- **Cleanliness- (2)** We believe that cleanliness may be an issue with this prototype. The issue comes down to the observed cleanliness of other tools within the Whitemarsh Special Education classroom. Possible remediations could include a 3D printed Clorox wipe holder.
- **Storage-(4)** We believe that this prototype will be very efficient in its storability. Due to the sinks' ability to be contained with a plastic container, as well as the ability to detach the water tanks, we believe that they will be able to be stored efficiently, without taking up a major part of the room.
- **Customizability (3)** - We believe this will receive a 3 due to the ability to use different sink handles, as well as implementing different mirrors to enhance the student experience.

Siphon Design

After reviewing the design further, we have decided that although this design works in theory the cons that come without the knowledge of siphons is the achilles heel of this design. The main pros that come with this design are the steady flow rate and the cleanliness. This is because a siphon has a steady flow rate when it flows leading to an optimal feature when it comes to a sink. The second stand out score is cleanliness, this is because the sink basin would be made out of stainless steel which after researching is one of the easiest materials to clean. Also the tubing is

inside of the faucet so when it comes to a child trying to mess with the interior tubing and getting it dirty is very unlikely. However the main downside of this design and the reason it is not the chosen design is the fact that we are not knowledgeable on the topic of making a siphon. Without this knowledge this task is going to take up a lot of time that we do not have,

Gravity Slope Design

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Cleanliness(6)	Use of materials that are easy to be cleaned with external products (acrylics, and wood, etc)	Inclusion of Soap container	Inclusion of Wet wipes (to wipe down sink)	A mini date-sheet, to remind the teachers of the last time cleaned, and approximately the next time the sink needs to be cleaned
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Scoring Rubric for Design Requirments

Design Objectives	Excellent (4)	Good (3)	Satisfactory (2)	Unacceptable (1)
Produce sufficient water flow (1)	>1 Gallons of Water per Minute	.5-.99 gallons of water per minute	.25-.49 gallons per minute	<.25 gallons per minute
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Storage(7)	10ft cubed or less	10.1-15 ft cubed	15.1-20 ft cubed	20.1-25 ft. cubed or more
Customizability(8)	3 or more parts of the sink can be modified for student/teacher use	2 parts of the sink can be modified for the student/teacher use	1 part of the sink can be modified for the student/teacher use	No parts of the sink can be modified for the student/teacher use

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Gravity Slope Design

Reflecting on this design, there are definitely some favorable pros, but a few costly cons. The main cons being the lack of portability and difficulty to design. This design would require us to build or purchase a large cabinet with a hole in the top. While it would be possible to build this, it would take much longer and we would need additional supplies/tools. When it comes to portability and storage, this design is much more bulky and awkward than the rest. The design may make it difficult for a teacher or teaching assistant to pick it up and bring it to a student's desk. It could benefit from the side handle bars the other designs utilize. A unique quality of this design is that the flow rate can easily be increased and adjusted with the 'automatic door esque' system. Overall, this design is hindered by bulky design not suitable for a kindergarten classroom. But, the detachable water storage and disposal system should be noted and potentially carried on to our final design.

Stainless-Steel sink basin	ORLANDO 13 x 13 inch Small Bar Sink Undermount Single Bowl Stainless Steel 18 Gauge Kitchen Sink With Basket Strainer - Amazon.com
Plastic Bins	https://www.amazon.com/CX-27-Gallon-Snap-Tight-Weather-Resistant-Organization/dp/B0C4VDJSKG/ref=sr_1_3_sspa?crid=RBZPAZF0Z35K&dib=eyJ2IjoiMSJ9.R33bVxmGY8GzZc3Ax6qTrumiTPs5R-QEnjnlQ6vY3wdUzOK89bUXg0wkGsKrU0dETuIzf4RRp0asx5e9rQ-cQPvUod7T2UsQWyVxbcu4xkyvQYG6OGkWbbgX_fPmw2gQcnhKQFGNvTABoPw9dOZx9p2AIs3T1TA8JIT8cHZ5a_8MX38hHZW_bqHXm9FULAwlq35IGhRtiJDa055s_kjgQT0Ic0EnBmAZpfiJS_D-ZbBY_oEIRAQUXCdsjesb0BWzeGAWwHoK_BhD402pWVNIrfWjZC0KEyuPPi7thPwIQ5o.im38WI d50yoKPVbtRta46hV3IVaCy9MAgO4Tdd8Q Qxo&dib_tag=se&keywords=plastic+container+black&qid=1705604804&srefix=plastic+container+bla%2Caps%2C123&sr=8-3-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&psc=1&smid=A24Y0KH77Z91NG
PVC	HYDROMAXX 1-1/2 in. x 25 ft. PVC Schedule 40 White Ultra Flexible Pipe 2201112025 - The Home Depot

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ote: The box that surrounds the tanks is not shown in order to display the system within it. The piping system was also not shown.

[illegible]

What is it	Link
Stainless-Steel sink basin	ORLANDO 13 x 13 inch Small Bar Sink Undermount Single Bowl Stainless Steel 18 Gauge Kitchen Sink With Basket Strainer - Amazon.com
Plastic Bins	https://www.amazon.com/CX-27-Gallon-Snap-Tight-Weather-Resistant-Organization/dp/B0C4VDJSKG/ref=sr_1_3_sspa?crid=RBZPAZF0Z35K&dib=eyJ2IjoiMSJ9.R33bVxmGY8GzZc3Ax6qTrumiTPs5R-QEnjnlQ6vY3wdUzOK89bUXg0wkGsKrU0dETuIzf4RRp0asx5e9rQ-cQPvUod7T2UsQWYVxbcu4xkyvQYG6OGkWbbgX_fPmw2gQcnhKQFGNvTABoPw9dOZx9p2AIs3T1TA8JIT8cHZ5a_8MX38hHZW_bqHXm9FULAwlq35IGhRtiJD055s_kjgQT0Ic0EnBmAZpfiJS_D-ZbBY_oEIRAQUXCdsjesb0BWzeGAWwHoK_BhD402pWVNlrfWjZC0KEyuPPi7thPwIQ5o.im38WI

	d50yoKPVbtRta46hV3lVaCy9MAgO4Tdd8QOxo&dib_tag=se&keywords=plastic+container+black&qid=1705604804&sprefix=plastic+container+bla%2Caps%2C123&sr=8-3-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&psc=1&smid=A24Y0KH77Z91NG
PVC	HYDROMAXX 1-1/2 in. x 25 ft. PVC Schedule 40 White Ultra Flexible Pipe 2201112025 - The Home Depot

Final Piece List

Link:	Quantity	Item	Cost	Estimated Weight
Plywood	1	Plywood 4x8 4608 480	\$30.62	61 total (will be trimmed down)
Sink Bowl -1515-6BS-1	1	Sink bowl	\$89.99	5.2
PVC Tubing	1	PVC Tubing	\$4.73	1.65
Plastic Tub	1	Plastic Tub	\$10.98	N/A
Water Storage	2	Water Storage	\$65.98	3.44

Link:	Quantity	Item	Cost	Estimated Weight
<i>Plywood</i>	<i>1</i>	<i>Plywood 4x8</i> <i>4608</i> <i>480</i>	<i>\$30.62</i>	<i>61 total</i> <i>(will be trimmed</i> <i>down)</i>
Container				
Dry-Wall Screws	<20	<i>Screws</i>	<i>\$7.28</i>	<i>N/A</i>
Extra tubing, tops, and faucet valves	1	Auxiliary tools in case of breakage	\$6.99	N/A
Total:			\$216.57	

Note that italicized materials were provided by the school's workshop inventory free of charge